Interactive Light and Media Installations for MAP mima

The Cube (Indoor Projections) & The Catenary (Outdoor LED)

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Introduction & Scoping Process

Search Strategy

In order to gather relevant inspiration for the MAP mima project, I conducted a structured search of two different spatial contexts: Immersive indoor environment ("Cube") and low-resolution outdoor public spaces ("The Catenary"). By accessing the ACM Digital Library and Google Scholar, I was able to find peer-reviewed papers in the field of human-computer interaction (HCI) in public spaces. I also search for real art projects through websites such as Designboom. My main keywords are "interactive projection", "public light art" and "sensor technology".

Source Reliability

To ensure the accuracy of this information, I visited the official websites of the artists (such as TeamLab and Reifik Andanor's website). These websites provided detailed explanations of the working principles of the sensors and lights. I also watched actual videos posted by visitors on YouTube and Vimeo to understand how people interacted with these artworks in practice.

Analysis Process

For each case study, I employed a three-step analysis method based on project evaluation criteria. First, I analyzed the user experience: I observed people's actual behaviors in the space (such as touching, moving, or sitting), and explained the psychological reasons behind these behaviors. Secondly, I analyzed the connection between technology and the place: I studied how sensors and lights served as a bridge to connect visitors with the physical room or outdoor area. Finally, I summarized the inspiration: I recorded the specific and practical ideas that our team could draw upon and apply to the MAP cube or catenary design.

Cube Case Study 1

Archive Dreaming by Refik Anadol

User Experience

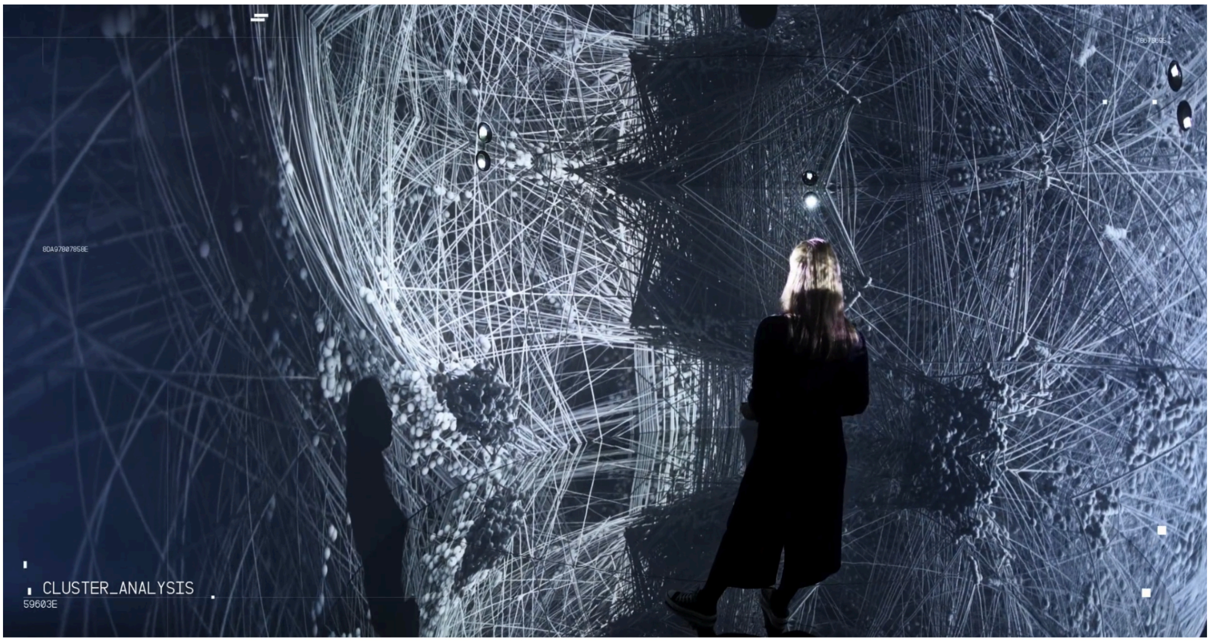
The "Archives Dreaming" project utilized millions of ancient historical documents to create a vast 360-degree panoramic video space. When people enter this room, those pictures and data flow around them like a whirlwind of information. This will bring about an extremely intense and astonishing experience. Because visual information is everywhere - covering the walls, ceilings and floors - people usually stop walking. They stood quietly in the center of the room, merely turning their bodies to look around. This made one feel as if one were inside a huge and beautiful digital brain, with oneself merely being a small traveler moving around within it.

Technology and Place Connection

This artist uses artificial intelligence technology to categorize the files and employs high-performance projectors to display the data throughout the room. This technology can completely conceal the real physical walls. It can transform an ordinary, dull square indoor space into a vast area with no clear boundaries. This technology enables users' thoughts to be directly connected to the digital space, thus making physical buildings appear soft and without boundaries.

Inspirations for The Cube

This project demonstrates that the "Cube" not only can play simple flat videos, but also can use computer algorithms to generate constantly changing dynamic patterns based on real-time data, thus achieving dynamic display. In our "MAP MiMa" project, this technology was adopted to achieve the dynamic effect. Furthermore, if we project light onto both the floor and the walls simultaneously, it will break the sense of a "box", making our small indoor space appear larger, more profound and more attractive.



Cube Case Study 2

Rain Room by Random International

User Experience

In the "Rain Room", people can walk through an intense indoor rainstorm, yet they remain completely dry. This is a truly extraordinary and completely different experience from reality. When people first see it, they often hesitate whether to enter or not. But once they step inside, they will become very curious and start testing the system. They moved slowly, extending their hands, or holding each other tightly, to see if the rain would reach them. This gave the tourists the feeling of having the power to control the weather, turning a simple walk into a fun performance.

Technology and Place Connection

The ceiling of the room is covered with 3D tracking cameras and thousands of water valves. When the cameras see a person moving below, they send a quick message to the computer to turn off the water valves directly above that person's head. The technology acts like an invisible, smart umbrella. It connects people to the room by creating a safe, dry space just for them, changing a harsh natural environment into a playful interactive zone.

Inspirations for The Cube

This case teaches us a great lesson about "negative space." An interaction does not always mean adding more light, more color, or more sound. Sometimes, we can take things away. For the MAP mima Cube, we can design a system where the projection on the wall or floor actually disappears or turns dark when someone walks close to it. This creates a personal, quiet shadow around the user.

Cube Case Study 3

Borderless by teamLab

User Experience

In the "Borderless" exhibition, digital natural elements such as flowers, animals and waterfalls can freely move on the exhibition walls. The most important thing is that if the user touches the flower on the wall, the petals of that flower will fall off and gradually wither. Because this artwork responds immediately to human touch, visitors will stay longer. Children and adults will naturally run around, touching the walls and observing what happens next. This gives people a strong sense of responsibility, as their actions directly change this beautiful environment.

Technology and Place Connection

This room is equipped with numerous infrared sensors and real-time computer graphics software. These sensors continuously scan the walls to precisely determine the exact location where people touch the walls. Subsequently, the computer processes the video in real time. As a result, people can establish a connection with that place because they no longer merely sit there passively watching the screen content. Their bodies and actions become a real part of the artwork's living ecosystem.

Inspirations for The Cube

Physical contact is a very effective way to attract users. For our "cube" design, we must allow and encourage people to touch its projected surface. We can use simple sensors. When users place their hands on the wall, the digital content will respond accordingly - for instance, water ripples will appear on the screen, or the colors around the fingers will change. In this way, this room becomes a vibrant playground.

Catenary Case Study 1

Waterlicht by Daan Roosegaarde

User Experience

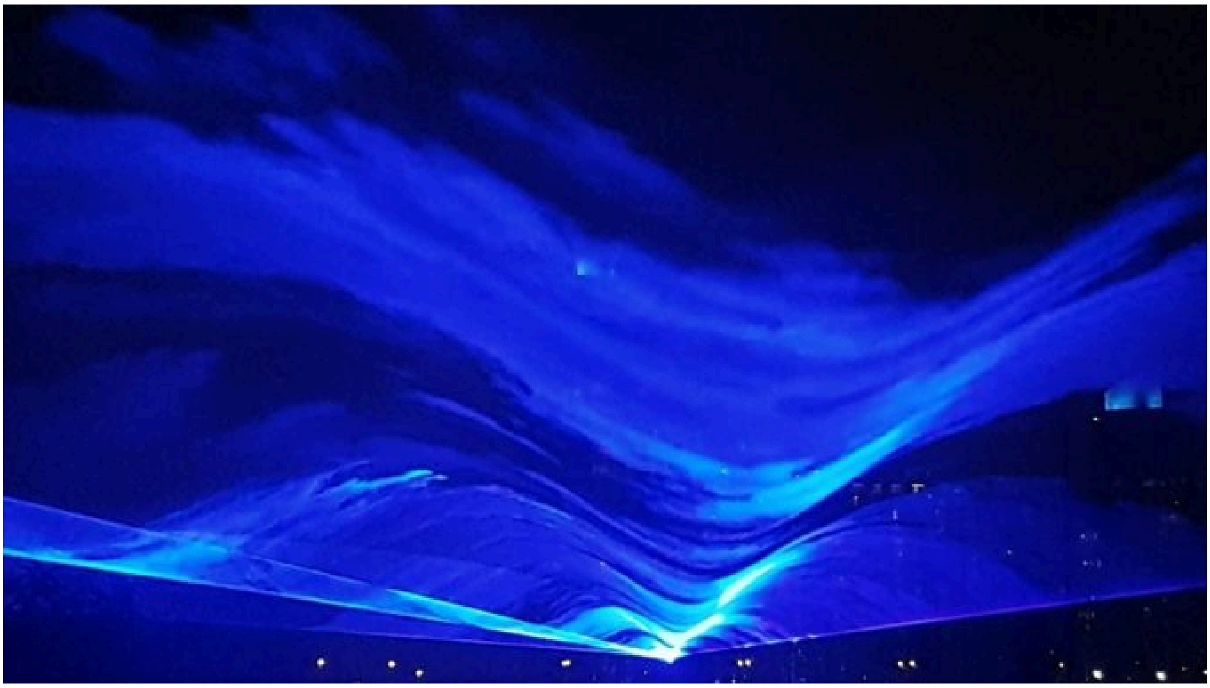
Waterlicht is an outdoor public artwork that uses strong blue laser lights to make it look like there is a huge flood of water floating just above people's heads. When walking into this outdoor square, people often slow down, and many choose to sit or lie on the ground to look up at the blue light. It creates an atmosphere that is both beautiful and very serious. It prompts people to deeply reflect on the issues of climate change and sea level rise, fostering a shared and emotional experience among strangers in this city.

Technology and Place Connection

This project employs powerful lasers, special lenses and smoke generators. Moreover, it heavily relies on the natural outdoor wind to drive the smoke flow. As a result, the laser beams appear to be moving waves on the sea surface. It ingeniously makes use of the spacious area at a high position (that is, the "catenary space"). By completely changing the way people perceive the height and history of urban squares, it closely connects people with the outdoor environment.

Inspirations for The Catenary

We can use the empty space above people's heads to build a "virtual roof of light". For the MAP MiMa chain design, we don't need to install shading screens on the ground. Instead, we can project the light horizontally. Furthermore, we can also utilize natural meteorological elements, such as natural wind or light rain, to interact with LED lights, making the outdoor installations appear more natural and full of vitality.



Catenary Case Study 2

Pulse Topology by Rafael Lozano-Hemmer

User Experience

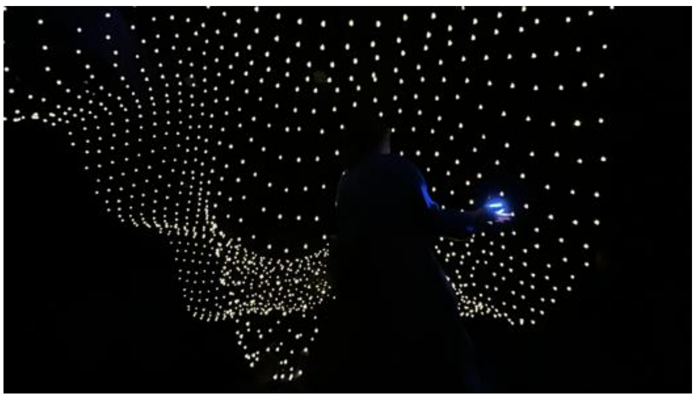
This outdoor installation consists of 3,000 light bulbs hanging in the air. The amazing part is that these lights flash exactly to the rhythm of the visitors' own heartbeats. This makes people feel incredibly special and noticed by the environment. Strangers often stop, hold the heartbeat sensors, and watch their personal rhythm travel through the lights. They start talking to each other, comparing their fast or slow heartbeats. It turns a very private body function into a beautiful, shared public game.

Technology and Place Connection

Visitors place their hands under special pulse sensors. These sensors read the blood flow in their fingers and send the heartbeat data to the hanging light bulbs above. The technology connects the human body directly to the outdoor architectural structure. It makes the cold, metal outdoor hanging lights feel warm, breathing, and alive, literally becoming the heartbeat of the public space.

Inspirations for The Catenary

This is a perfect example of using biological data for outdoor installations. Instead of asking people to use a mobile phone or a complicated platform, we can use simple sensors to collect body data. For MAP mima, we can measure how fast the crowd is walking or use hand sensors, and then use that simple data to change the brightness or speed of the Catenary LED lights in real-time.



Catenary Case Study 3

The Bay Lights by Leo Villareal

User Experience

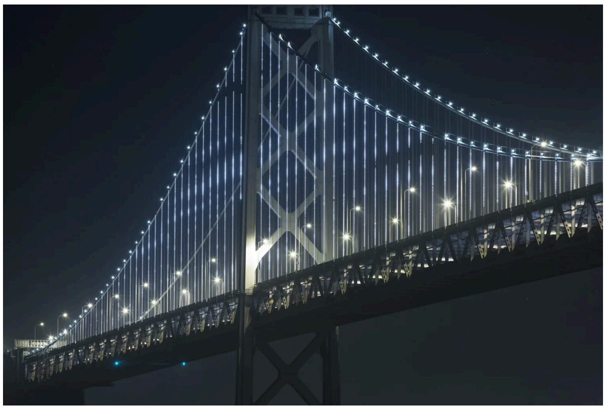
The Bay Lights is a massive art piece installed on a huge bridge in San Francisco. It uses 25,000 simple white LED lights. It does not show clear pictures, text, or movies. Instead, it shows abstract, moving light patterns that look like ocean waves or rain. People look at it from a distance while driving or walking home. It does not demand people's full attention or make them tired. Instead, it provides a very relaxing, calming, and continuous visual rhythm for the city at night.

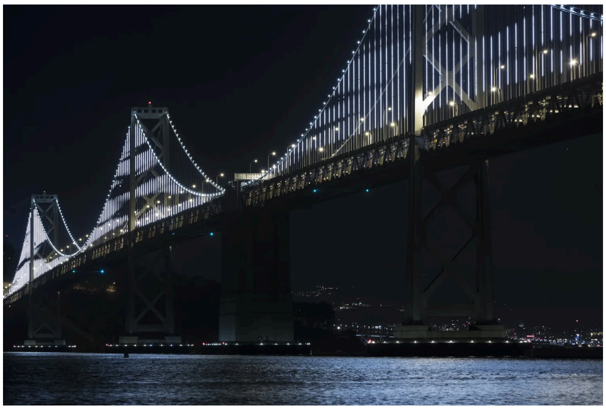
Technology and Place Connection

A complex computer algorithm controls the simple LED lights to make random, moving patterns that never repeat exactly the same way. The technology fits perfectly with the massive scale of the outdoor bridge. It does not try to turn the bridge into a giant, annoying TV screen. Instead, it respects the public space and works together with the bridge's cables to make the night view more elegant and peaceful.

Inspirations for The Catenary

This shows that outdoor Catenary designs do not need high-resolution or expensive screens. Simple, low-resolution LED lights are actually better for large outdoor areas. For MAP mima, we should not try to show complex graphics on the outdoor lights. Instead, we should focus on making smooth, abstract, and slow-changing light patterns that match the peaceful outdoor environment.





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